

Youssef Naciri Zerta

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EMBL-EBI, Hinxton — Saez Lab (2025–) | Research thesis at EMBL-EBI, February–August 2027 | PhD applications, autumn 2027

RESEARCH FOCUS

The question I work on is **control of cell state**: given a cell in one state and a target state, can an intervention that moves it there be *derived* rather than found by screening? That is not a screening problem with more throughput — it requires knowing what a cell's state actually is, which interventions carry information about it, and at what level of organisation causal relationships become predictable rather than merely descriptive. The transcriptome may be informative without being a sufficient description of the state that determines a cell's response, and genes may be real entities without being the coordinates in which control is naturally expressed.

My current work attacks the first obstacle in the way of that: whether the interventions perturbational genomics actually performs can identify the structure the field claims to recover from them. The methods I bring to it are causal representation learning and identifiability theory, mechanistic and dynamical modelling, perturbational genomics, and infrastructure that makes causal inference over biological structure comparable and composable — tools for the question rather than the subject of it. The recurring result so far is negative: representational claims tend not to survive contact with interventional ground truth, and the failure is usually in the null a claim is read against rather than in the model.

RESEARCH

Identifiability of Gene-Level Perturbation

2026–2027

EMBL-EBI, Saez Lab — with Rory Maizels (Saez Lab, EMBL-EBI; Parts Group, Sanger; Briscoe Group, The Crick) – research project 09/2026–02/2027 (remote, part time); research thesis 02/2027–08/2027 (full time, on site)

- Perturbation biology assumes that transcriptomic responses to gene-level interventions are an adequate basis for learning regulatory mechanism. I test that from the identifiability side: which aspects of a regulatory system can in principle be recovered from the interventions actually performed, where information is fundamentally missing, and whether apparent model performance survives systems with known causal ground truth.
- Building ODE (ordinary differential equation) simulators of published GRN (gene regulatory network) models spanning monostable, multistable and oscillatory dynamics, with a count-sampling observation model; five intervention classes — noisy knockdown, perfect knockout, edge addition, edge removal, edge tweak — of which only the first two are node interventions expressible as $\text{do}(\cdot)$ on observed variables.
- Bayesian parameter inference per intervention class, scored on structure recovery, parameter recovery and held-out prediction against ground truth; the failure signatures found in silico are then tested in public Perturb-seq data (Replogle genome-scale CRISPRi, Norman CRISPRa).
- The second phase moves from node-level to calibrated mechanism-level interventions measured in cells, where what is delivered is a function of cell state rather than the identity of a gene — the first step toward deriving an intervention rather than selecting one from a screen.

Sparsity, Not Alignment: How a Standard Interpretability Test Manufactures Discovery

2026

Sole author — under review, NeurIPS 2026 workshop – *sparse autoencoders in a genomic foundation model, audited against causal ground truth*

- Built a causally grounded test of whether SAE (sparse autoencoder) features of a genomic foundation model correspond to regulatory mechanism, using saturation-mutagenesis MPRA (massively parallel reporter assay) as per-base-pair ground truth. Applied directly it reports a discovery that passes every control the literature asks for — label shuffling, repeat masking, a likelihood baseline, three seeds, two autoencoder families, two architectures.
- The discovery is an artifact. The procedure thresholds one tail of a null and never checks that the excess is one-sided: it is not, the same FDR (false discovery rate) procedure on the opposite tail returns more discoveries, the distributions differ in scale rather than location, and the effect grows with dictionary size. What the threshold selects is units whose score is more variable than a dense random direction's — which sparser units mechanically are.
- Four diagnostics that expose this at no extra cost, a studentised repair, and a noise ceiling showing the accompanying reproducibility failure is a property of the ground-truth budget rather than of the decomposition. The failure mode is generic to any pipeline that thresholds a noisy per-unit statistic.

AnnNet — Lead Developer

2025–Present

EMBL-EBI, Saez Lab, with Open Targets – *annotated network structure for systems biology, and the method ecosystem built on it; co-first-author preprint in preparation (Bioinformatics)*

- **AnnNet** is AnnData for systems biology. One annotated container holds directed, signed, hyperedge, multilayer and condition-specific structure on a single sparse incidence core, with typed annotation tables, lossless native storage and lazy interoperability with NetworkX, igraph, graph-tool and PyTorch Geometric — the classical directed graph is a special case rather than the frame.
- I designed and built the core: the incidence representation, the annotation model, the storage format and the library interfaces. Used by groups at EMBL-EBI and Imperial College London, and developed in close contact with scverse core developers and collaborators, so that it composes with the single-cell stack rather than competing with it.
- SysBioVerse (2026–, Open Targets) is the layer on top: the methods, and the connectors to the prior-knowledge ecosystem, that make AnnNet biology-aware. Regulatory, signalling, metabolic, cell-cell and multi-omic causal inference behind one interface, so that methods which disagree about what a causal biological structure even is can still be compared, benchmarked and composed rather than reimplemented per paper against incompatible data structures.
- The motivation is the same one that drives the rest of my work: biological structure is routinely multilayer and context-dependent, and forcing it into a single directed graph discards exactly the conditioning that makes a network claim causal.

SampleCLR — Patient-Level Representations from Single-Cell Transcriptomics

2025

Helmholtz Munich, Lücken Lab – *contrastive and attention-based sample representation* [preprint]

- Contributed interpretability and transfer-learning analyses, and to the architecture design, for a model producing patient-level representations from single-cell data with state-of-the-art disease-stratification performance.
- Evaluated whether the representations transferred across biological systems rather than only fitting the training benchmark — the question my later work formalises.

Multiplex Bacterial Interaction Networks

2024–Present

Université Euromed de Fès – *experimental systems biology; first author; in preparation (ISME J)*

- Designed and executed a trait-resolved interaction study across 60 soil bacterial strains, generating 12 directed interaction layers from pairwise assays — an interaction tensor rather than a single network.
- Built a multilayer-network analysis of that tensor to characterise microbial regulatory strategies across ecological contexts, showing that interaction sign and strength are context-dependent rather than intrinsic to a strain pair — the limitation that motivates AnnNet: a single directed graph cannot represent a system whose edges change meaning with condition.

Also: analysis pipeline for detecting biomarker combinations from epigenomic data, MD Anderson Cancer Center / KAUST, 2024–2025.

PUBLICATIONS & PREPRINTS

- **Y. Naciri Zerta.** *Sparsity, not alignment: how a standard interpretability test manufactures discovery in a genomic foundation model.* Sole author, under review, NeurIPS 2026 workshop. [earlier preprint]
- *AnnNet: a unified annotated-graph structure for biological networks.* Co-first author, in preparation (Bioinformatics).
- *Patterns of pairwise interactions between Streptomyces spp.* First author, in preparation (ISME J).
- *Great patients embed alike: contrastive learning for sample representation from single-cell data.* Middle author. [preprint]

EDUCATION

Ingénieur d'État en Biotechnologies (State Engineer, Biotechnology)

Expected 2027

Université Euromed de Fès (UEMF), Morocco – *five-year engineering degree; Master's-equivalent, 300 ECTS (European Credit Transfer and Accumulation System)*

Relevant coursework: Systems Biology, Synthetic Microbiology, Recombinant DNA Technology, Statistical Genetics, Analytic Chemistry, Organic Chemistry, Probability & Statistics.

METHODS

Causal & representation learning: causal representation learning, identifiability theory, interventional and counterfactual inference, sparse and overcomplete latent-variable models, mechanistic interpretability, foundation models.

Statistical modelling: high-dimensional and sparse inference, Bayesian and simulation-based inference (MCMC, SMC-ABC), spectral methods, stochastic dynamics and linear-response theory.

Evaluation design: interventional ground truth, null calibration and misspecification diagnostics, permutation and subsampling inference, multiple testing, group-structured cross-validation, leakage diagnosis, floor and ceiling calibration, planted nulls, pre-registration.

Genomics & regulation: gene regulatory networks, perturbational genomics, single-cell transcriptomics, saturation

mutagenesis and genome editing, statistical genetics.

Systems biology: mechanistic modelling, dynamical systems, multilayer and hypergraph formalisms, prior-knowledge integration.

Engineering: Python/PyTorch, R, C/C++, scientific library design, open-source maintenance, GPU training, cloud computing, Docker.

Wet lab: molecular cloning, genome engineering, microbial culture, analytic chemistry.

Languages: English & French (fluent), Arabic & Tamazight (native), Spanish (basic).